The Solutions to Exam 2025

B—bookwork, A—application, E—new example, T—new theory

1.

a) $H(X) = -\Sigma\, P(x) \log_2 P(x)$ [2E]

$= -[0.5 \log_2 0.5 + 0.3 \log_2 0.3 + 0.2 \log_2 0.2]$ [2E]

≈ 1.485 bits [2E]

b) $H(X) = -[0.5 \log_2 0.5 + 0.5 \log_2 0.5] = 1$ bit [1E]

$H(Y) = -[0.3 \log_2 0.3 + 0.7 \log_2 0.7] \approx 0.881$ bits [1E]

$H(X,Y) = -\Sigma\, P(x,y) \log_2 P(x,y) \approx 1.846$ bits [2E]

$I(X;Y) = H(X) + H(Y) - H(X,Y) \approx 1 + 0.881 - 1.846 \approx 0.035$ bits [2E]

c) $D(P \,||\, Q) = \Sigma\, P(x) \log_2 [P(x) / Q(x)]$ [2E]

$= 0.5 \log_2 [0.5 / 0.4] + 0.3 \log_2 [0.3 / 0.4] + 0.2 \log_2 [0.2 / 0.2]$ [2E]

$\approx 0.5(0.32) + 0.3(-0.415) + 0$

$\approx 0.16 - 0.125 \approx 0.035$ bits [2E]

d) Fano's inequality:

$H(X|Y) \leq H(Pe) + Pe \log_2(|X| - 1)$ [2E]

$H(Pe) = -[0.1 \log_2 0.1 + 0.9 \log_2 0.9]$

$\approx -[0.1(-3.32) + 0.9(-0.152)]$

$\approx 0.332 + 0.137 \approx 0.469$ bits [2E]

$H(X|Y) \leq 0.469 + 0.1 \log_2(2 - 1)$

$= 0.469 + 0.1(0) = 0.469$ bits [3E]

Note: if student uses the bound $H(X|Y) \leq H(Pe) + Pe \log_2(|X|)$, it is also correct. Then the answer will be

$H(X|Y) \leq 0.469 + 0.1 \log_2(2)$

$= 0.469 + 0.1(1) = 0.569$ bits

2.

a)

(1) Send nothing when $D \geq p$, as the distortion is p. [1B]

(2) definition of mutual information [1B]

(3) $H(x) = H(p)$

$H(x|\hat{x}) = H(x \oplus \hat{x} | \hat{x})$ [1B]

(4) conditioning reduces entropy [1B]

(5) $H(x \oplus \hat{x}) \leq H(D)$ [3B]

This is because $E_{x,\hat{x}} d(x, \hat{x}) \leq D$

$= 1 \cdot Pr(x \oplus \hat{x} = 1) + 0 \cdot (Pr\ x \oplus \hat{x} = 0)$

$= Pr(x \oplus \hat{x} = 1) \leq D$

$H(x \oplus \hat{x}) \leq H(D)$ as $H(p)$ is monotonic when $p \leq \frac{1}{2}$

(6) $I(x; \hat{x}) \geq H(p) - H(D)$ [2B]

$\Rightarrow \min I(x; \hat{x}) \geq H(p) - H(D)$

$R(D) \geq H(p) - H(D)$

b)

i) This follows from the test channel.

$R(D) = \min I(x; \hat{x}) = \min h(x) - h(x|\hat{x})$ [3A]

$= \min h(x) - h(z) \geq h(x) - \frac{1}{2}\log(2\pi e D)$ [3A]

since Gaussian distribution maximizes the entropy.

ii)

By definition

$$R(D) = \min I(X;\hat{X})$$

Such that $E[(X-\hat{X})^2] = D$ and $p(x) = \int p(x,\hat{x})\,d\hat{x}$. [2T]

The second condition is obviously true from Fig 2.2.

Now check the first condition:

$$E[(X-\hat{X})^2] = E\left[\left(\frac{D}{\sigma^2}X - \frac{\sigma^2-D}{\sigma^2}Z\right)^2\right]$$

$$= \frac{D^2}{\sigma^4}\sigma^2 + \frac{(\sigma^2-D)^2}{\sigma^4}\frac{D\sigma^2}{\sigma^2-D^2} \quad [2T]$$

$$= \frac{D^2}{\sigma^2} + \frac{(\sigma^2-D)D}{\sigma^2}$$

$$= D.$$

The mutual information

$$I(X;\hat{X}) = h(\hat{X}) - h(\hat{X}|X)$$

$$= h(\hat{X}) - h\left(\frac{\sigma^2-D}{\sigma^2}Z\right) \quad [2T]$$

Note that $\hat{X}$ has zero mean and variance

$$E[\hat{X}^2] = \left(\frac{\sigma^2-D}{\sigma^2}\right)^2\left(\sigma^2 + \frac{\sigma^2 D}{\sigma^2-D}\right) \quad [2T]$$

$$= \sigma^2 - D$$

So $h(\hat{X}) \leq \frac{1}{2}\log[2\pi e(\sigma^2-D)]$ Gaussian has maximum entropy

Hence,

$$I(X;\hat{X}) \leq \frac{1}{2}\log\left[2\pi e(\sigma^2 - D)\right] - h(Z) - \log\frac{\sigma^2 - D}{\sigma^2}$$

$$= \frac{1}{2}\log\left[2\pi e(\sigma^2 - D)\right] - \frac{1}{2}\log\left[2\pi e\frac{D\sigma^2}{\sigma^2 - D}\right] - \frac{1}{2}\log\left(\frac{\sigma^2 - D}{\sigma^2}\right)^2$$

$$= \frac{1}{2}\log\frac{\sigma^2}{D} \qquad [2\text{T}]$$

Since $I(X;\hat{X}) \leq \frac{1}{2}\log\frac{\sigma^2}{D}$ for this example, the minimum mutual information also $\leq \frac{1}{2}\log\frac{\sigma^2}{D}$. QED.

3.

a)

i) Weakly symmetric channel [3E]

$$C = \log(3) - H\left(\frac{1}{2}, \frac{1}{3}, \frac{1}{6}\right) = 0.126$$

ii) Symmetric channel [3E]

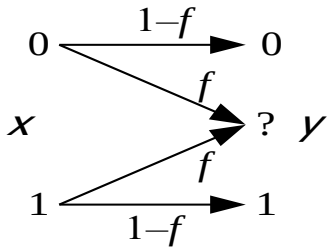
$$C = \log(3) - H\left(\frac{1}{2}, \frac{1}{3}, \frac{1}{6}\right) = 0.126$$

iii) Symmetric channel [3E]

$$C = \log(4) - H\left(\frac{1}{2}, \frac{1}{3}, \frac{1}{6}\right) = 0.541$$

b)

(i) The transition matrix of a BEC



can be rearranged into

$$\begin{array}{c} \\ 0 \\ 1 \end{array} \begin{array}{c} \begin{array}{ccc} 0 & 1 & ? \end{array} \\ \left(\begin{array}{cc:c} 1-f & 0 & f \\ 0 & 1-f & f \end{array}\right) \end{array}$$

Partition this matrix into two blocks, and note that both blocks are symmetric. So this is a generally symmetric channel. [3T]

However, BEC is not a weakly symmetric channel, because the column sums are not identical in general (unless f = 1/3). [1A]

(ii)

The transition matrix of the first channel is given by

$$\begin{pmatrix} 0.7 & 0.2 & 0.1 \\ 0.1 & 0.2 & 0.7 \end{pmatrix}$$

which can be rearranged into [1T]

$$\left(\begin{array}{cc:c} 0.7 & 0.1 & 0.2 \\ 0.1 & 0.7 & 0.2 \end{array}\right)$$

Again, with the above partition, both blocks are symmetric. So this channel is generally symmetric. We can still use the formula

$$I(X;Y) = H(Y) - H(Y \mid X) = H(Y) - H(\mathbf{Q}_{1,:})$$

to calculate capacity, with uniform input distribution (because it achieves capacity). The difference here is that the output distribution is not uniform anymore. Thus, [2T]

$$C = H(0.4, 0.4, 0.2) - H(0.7, 0.2, 0.1) = 1.52 - 1.16 = 0.36 \text{ bits}$$

The transition matrix of the second channel is given by

$$\begin{pmatrix} 0.7 & 0.2 & 0.1 \\ 0.2 & 0.1 & 0.7 \end{pmatrix}$$

There is no way to do a similar partition, so it is not generally symmetric. [1T]

c)

(i) Recall the definition of mutual information

$$I(X;Y) = \sum_y P(y|x)Q(x) \log \frac{P(y|x)}{\sum_x Q(x)P(y|x)} \quad (*)$$

Finding the capacity can be reformulated as the following optimization problem

$$\max_{Q(x)} I(X;Y) \qquad \text{subject to} \quad \sum_x Q(x) = 1$$

Using the method of Lagrange multiplier, we form the objective function

$$J = I(X;Y) + \lambda \sum_x Q(x)$$

Taking partial derivative with respect to Q(x), we obtain [2T]

$$\frac{\partial J}{\partial Q(x)} = I(x;Y) - \log e - \lambda$$

Therefore, letting this partial derivative be 0, we have

$$I(x;Y) = constant = C$$

Substituting this back to Eq. (*), we have [2T]

$$\max_{Q(x)} I(X;Y) = C$$

(ii)

For a generally symmetric channel, if the input distribution is uniform, then

$$I(x;Y) = \sum_y P(y|x) \log \frac{P(y|x)}{\sum_x \frac{1}{|X|} P(y|x)}$$

Note that within each symmetric block of a partition, $\sum_x \frac{1}{|X|} P(y|x)$ is identical for all y's, because its columns are permutations of each other. [2T]

This implies that if we form a matrix of entries

$$P(y|x) \log \frac{P(y|x)}{\sum_x \frac{1}{|X|} P(y|x)}$$

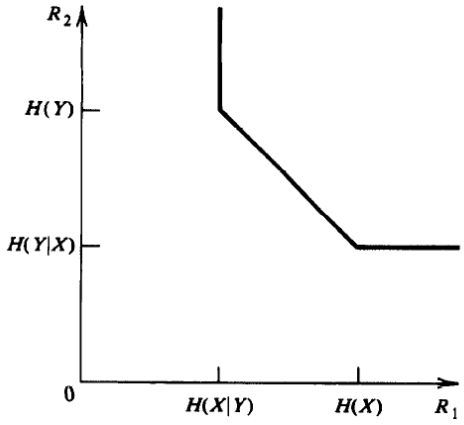
its rows will be permutations of each other. Thus $I(x;Y)$ is a constant for all x's. Therefore, the condition in (i) is satisfied, and accordingly the uniform distribution achieves capacity. [2T]

4.

a)

The Slepian-Wolf region is given by

$$R_1 \geq H(X \mid Y)$$
$$R_2 \geq H(Y \mid X)$$
$$R_1 + R_2 \geq H(X,Y)$$



[4E]

In this question,

$$H(x) = -\frac{3}{8}\log\frac{3}{8} - \frac{5}{8}\log\frac{5}{8} = 0.954 \text{ bits}$$

$$H(y) = -\frac{1}{2}\log\frac{1}{2} - \frac{1}{2}\log\frac{1}{2} = 1 \text{ bit}$$

$$H(x,y) = -\tfrac{1}{4}\log\tfrac{1}{4} - \tfrac{1}{4}\log\tfrac{1}{4} - \tfrac{1}{8}\log\tfrac{1}{8} - \tfrac{3}{8}\log\tfrac{3}{8} = 1.906 \text{ bits} \quad \text{[2E]}$$

$$I(x;y) = H(x) + H(y) - H(x,y) = 0.048 \text{ bits}$$

$$H(y|x) = H(y) - I(x;y) = 0.952 \text{ bits} \quad \text{[2E]}$$

$$H(x|y) = H(x) - I(x;y) = 0.906 \text{ bits} \quad \text{[2E]}$$

b)

i) The capacity of erasure channel from X to Y1 is given by

$$C_1 = 1 - \alpha_1$$

[5E]

ii) Note that the channel from X to Y2 is an equivalent erasure channel with parameter

$$\alpha_1 + \alpha_2 - \alpha_1\alpha_2$$

[5E]

Thus, the capacity of the channel from X to Y2 is given by

$$C_1 = 1 - \alpha_1 - \alpha_2 + \alpha_1\alpha_2 = (1 - \alpha_1)(1 - \alpha_2)$$

[5E]